Mathematical Modeling

Four mock labs: standard form and simplex

Personal study notes

How to use these tasks

Complete the solved demo first. Then attempt each mock lab before opening the separate answer PDF or its reference MATLAB script. The time budgets below are suggested practice limits. Each task is independent. For hand work, standard form means $Sv = b$, $v \geq 0$, with a maximization objective. Convert minimization by negating the objective. For MATLAB, use `linprog` with `dual-simplex-highs`; remember that it minimizes. Put variable bounds into matrix rows and pass empty `lb`, `ub`.

In every submission, include variables and units, the original model, the standard-form matrix and variable order, a starting-basis check, the requested hand work, MATLAB arrays and output, and an interpretation in the original variables. Check solver status, signs, constraint residuals and the objective. Use 10^{-7} as the absolute comparison tolerance. Do not require identical optimal vectors or split-variable values when the representation is nonunique.

For hand pivots use the most negative $z_j - c_j$, breaking entering ties by smaller column index. Break leaving-ratio ties by the smaller basic-variable index. Keep variable headings fixed. Include c_B, z_j, and $z_j - c_j$; place slash ratios and the leaving arrow outside the RHS, as in the demo.

1 Mock lab A: a feasible slack start (40 minutes)

A workshop makes two divisible products. Product 1 earns 5 monetary units per batch and uses 3 hours of assembly and 1 hour of finishing. Product 2 earns 4 per batch and uses 2 hours of each resource. There are 18 assembly hours and 10 finishing hours. Unsold inventory is not required and production may be zero.

1. Formulate the profit-maximizing LP. State the dimensions of all arrays.
2. Add slacks s_1, s_2 in that order. Use the fixed column order (x_1, x_2, s_1, s_2). Identify and verify an initial BFS.
3. Solve by hand, showing every tableau. Explain the stopping condition.
4. Verify both the original inequalities and standard-form equalities with MATLAB. Report production, unused hours and total profit.
5. Draw the feasible polygon and mark your hand pivot path.
6. Repeat the solver calculation with 20 assembly hours and unchanged finishing hours. Explain the change in production and profit.

Suggested self-marking: model 2, standard form/start 2, pivots 3, MATLAB checks 2, interpretation 1 (10 points total).

2 Mock lab B: a start that needs Phase I (50 minutes)

Two divisible feed components cost 2 and 3 monetary units per kg. Their masses are $x_1, x_2 \geq 0$. At least 4 kg of feed is required. A processing condition requires exactly $x_1 + 3x_2 = 6$. Minimize total cost.

1. Write the original LP. Explain why subtracting a surplus variable from the minimum-mass constraint does not immediately give a feasible slack/surplus basis.
2. Use the original standard-form order (x_1, x_2, s_1). Add artificials a_1, a_2 in that order for the hand calculation, giving $(x_1, x_2, s_1, a_1, a_2)$.
3. Start with (a_1, a_2) basic and maximize $-a_1 - a_2$. Construct the indicator row from the actual basic coefficients. Carry out Phase I and state what its optimum proves.
4. Remove artificial columns when justified. Recompute the original objective as a maximization of negative cost and determine whether a Phase II pivot is required.
5. Solve the original LP with MATLAB, then solve its original three-variable standard form without artificial variables. Explain why the solver input does not require your hand Phase I variables.
6. Independently eliminate one variable using the equality and minimize the resulting one-variable expression on its feasible interval.

Suggested self-marking: model/standard form 2, Phase I 4, Phase II 2, solver and independent check 2 (10 points total).

Preparation note

This is an extension after the slack-start task. Read Chapter 3's initialization section before attempting it if Phase I is unfamiliar.

3 Mock lab C: recover the original variables (45 minutes)

A continuous planning model has a signed adjustment u with no sign restriction, and a second adjustment v satisfying $-1 \le v \le 2$. Maximize the score

$$F(u, v) = 3u + 2v + 5$$

subject to

$$-u - v \ge -3, \qquad -u + v \le 1.$$

The score and variables are abstract numerical quantities; no integrality is assumed.

1. Convert the first inequality to a $\le$ row, explaining the sign reversal. In the original variables, assemble all restrictions into $A(u, v)^T \le b$. Do not accidentally impose $u \ge 0$ or $v \ge 0$.

2. Write $u = u^+ - u^-$ and $v = w - 1$, with $u^+, u^-, w \ge 0$. Derive the new RHS values, the upper bound on w, the objective coefficients, and the objective constant.

3. Add slacks s_1, s_2, s_3 and use the fixed standard-form order $(u^+, u^-, w, s_1, s_2, s_3)$. Verify a feasible slack basis and calculate its meaning in the original variables.

4. Perform the improving hand pivot(s). Recover u, v and the full score, including the objective constant.

5. Solve both formulations with MATLAB. Compare recovered original decisions and full scores. Must the solver return exactly the same individual values of u^+ and u^- as your hand calculation?

6. Give a direct upper bound on F using $u + v \le 3$ and $v \ge -1$.

Suggested self-marking: signs/domains 2, transformation 3, simplex 2, recovery and solver comparison 2, certificate 1 (10 points total).

4 Mock lab D: diagnose the outcome (40 minutes)

For each model below, classify the mathematical outcome before interpreting any solver output. All variables are continuous and nonnegative.

D1: a finite maximum

$$\max(x_1 + x_2), \qquad x_1 + x_2 \le 4, \quad x_1 \le 3, \quad x_2 \le 3.$$

Convert to standard form using three slacks. Find the optimal value, give two distinct optimal points, and describe every optimizer. Verify the model with MATLAB. Explain why a different returned vector need not be an error.

D2: competing requirements

$$\min(x_1 + 2x_2), \qquad x_1 + x_2 \ge 5, \quad 2x_1 + 2x_2 \le 8.$$

Give a short mathematical feasibility certificate. For an additional Phase I check, use

$$x_1 + x_2 - s + a = 5, \qquad 2x_1 + 2x_2 + t = 8, \qquad x_1, x_2, s, t, a \ge 0.$$

Determine the minimum possible artificial sum a and exhibit a point attaining it. Compare with the solver status for the original model. Explain why no original optimal decision vector should be reported.

D3: an improving direction

$$\max(2x_1 + x_2), \qquad x_1 - x_2 \le 1.$$

Write standard form with one slack. Carry out the first pivot from the slack basis and inspect the next entering column. Give a feasible point and a nonnegative improving ray, verify the constraint along it, and compute the objective as a function of its ray parameter. Compare with the solver status.

Suggested self-marking: D1 3, D2 3, D3 4 (10 points total).

Sources and editorial remarks

All four tasks, suggested times and marking guides are invented for practice; they are not official Lab 3 assignments. The confirmed Lab 3 topics are simplex and standard form. Assessment of Phase I, free-variable transformations or a particular MATLAB implementation remains unconfirmed. Tasks B–D deliberately broaden practice. The exact lab timetable remains unconfirmed. The original Lab 3 starter file is left untouched.